

Bachelor's Thesis Assignment



156402

Institut: Department of Intelligent Systems (DITS)
Student: **Hucovič Vladimír**
Programme: Information Technology
Title: **Efficient Instrumentation of Programs using LLVM Framework**
Category: Software analysis and testing
Academic year: 2023/24

Assignment:

1. Get acquainted with the Perun project (performance version system) and the field of software profiling.
2. Study the LLVM infrastructure and its program instrumentation methods for resource consumption measurement (e.g., function duration, executed instructions or memory consumption).
3. Design and implement a profiler using LLVM-based instrumentation. Focus on improving the efficiency and/or precision of the instrumentation and resource measurement processes.
4. Design and implement suitable visualisation of the resulting collected data (e.g., flame graph or tree view), or use and enhance one of the existing visualisations in Perun.
5. Demonstrate the solution on at least one non-trivial use-case.

Literature:

- Oficiální stránky projektu Perun: <https://github.com/Perfexionists/perun>
- The LLVM Compiler Infrastructure: <https://llvm.org/>
- Gregg, B. (2020). Systems Performance, (2nd ed.). Pearson. ISBN: 9780136821694.

Requirements for the semestral defence:

First two points of the assignment.

Detailed formal requirements can be found at <https://www.fit.vut.cz/study/theses/>

Supervisor: **Pavela Jiří, Ing.**
Consultant: Fiedor Tomáš, Ing., Ph.D.
Head of Department: Hanáček Petr, doc. Dr. Ing.
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